

Workbook: Tissues and Histology

Week 1 workbook · BIO 004 Human Anatomy · Dr. Sharilyn Rennie · Solano Community College

A comprehensive workbook on the four tissue types, the connective tissue plan, the histology of each tissue and its subtypes, body membranes, and tissue repair. Work each question on your own first, then click *Reveal* to check your answer. The questions progress from recall to comparison to clinical reasoning, grouped by topic. Print the question-only PDF if you want to write longhand.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

Tissue origins and overview · 6 questions

Q01 · DOK 1 · Recall

Define histology and define tissue. How are the two terms related?

Q02 · DOK 1 · Recall

Name the three primary germ layers and one major tissue or structure each gives rise to.

Q03 · DOK 1 · Recall

Place the three germ layers in order from outermost to innermost as they sit in the early embryo.

Q04 • DOK 1 • Recall

List the four primary tissue types and the primary function of each.

Q05 • DOK 1 • Recall

For each of the four tissue types, name one location where it dominates.

Q06 • DOK 2 • Compare

Rank the four tissue types by how much extracellular matrix they contain, from most to least, and explain why the ranking makes sense.

Epithelial tissue: general features • 9 questions**Q07 • DOK 1 • Recall**

List the five major functions of epithelial tissue.

Q08 • DOK 1 • Recall

List the general structural features that define epithelial tissue.

Q09 · DOK 2 · Apply

Explain what polarity means in an epithelial cell and why it matters for function.

Q10 · DOK 1 · Recall

What is the basement membrane, what does it sit between, and what does it do?

Q11 · DOK 2 · Apply

Epithelial tissue is avascular. How does it still receive oxygen and nutrients, and what is the trade-off?

Q12 · DOK 2 · Apply

Epithelia regenerate well. Why is that an advantage given where epithelia sit in the body?

Q13 · DOK 1 · Recall

Epithelia are classified on two main axes. What are they?

Q14 · DOK 2 · Apply

Compare simple and stratified epithelia by structure and the kind of job each is built for.

Q15 · DOK 2 · Compare

Compare squamous, cuboidal, and columnar cells by shape and the job that shape suits.

Epithelial tissue: named types and locations · 7 questions**Q16 · DOK 1 · Recall**

Where is simple squamous epithelium found, and what is it doing there?

Q17 · DOK 1 · Recall

Where is simple cuboidal epithelium found?

Q18 · DOK 1 · Recall

Where is simple columnar epithelium found, and what specialized cells often appear within it?

Q19 • DOK 2 • Apply

What is pseudostratified epithelium, why does it look layered when it is not, and where in the body is it most important?

Q20 • DOK 1 • Recall

Compare keratinized and non-keratinized stratified squamous epithelium by structure and by where they are found.

Q21 • DOK 2 • Apply

What is transitional epithelium, where is it found, and what is the special property of its surface cells?

Q22 • DOK 1 • Recall

Define endothelium and mesothelium. Where is each found?

Epithelial junctions • 7 questions**Q23 • DOK 1 • Recall**

Name the major types of cell-cell junctions in epithelial tissue and the job of each.

Q24 · DOK 1 · Recall

What does a hemidesmosome do, and how does that differ from a desmosome?

Q25 · DOK 1 · Recall

What proteins make up a tight junction, and what does each contribute?

Q26 · DOK 1 · Recall

What proteins make up an adherens junction, and what is the junction anchored to inside the cell?

Q27 · DOK 1 · Recall

What proteins make up a desmosome, and what is the desmosome anchored to inside the cell?

Q28 · DOK 1 · Recall

What proteins make up a hemidesmosome, and what does it attach the cell to?

Q29 · DOK 1 · Recall

What proteins make up a gap junction, and how are they organized to form the channel?

Cilia, flagella, and apical surface specializations · 10 questions

Q30 · DOK 1 · Recall

What are microvilli, where are they found, and what is their function?

Q31 · DOK 1 · Recall

What are cilia, where are they prominent, and what do they do?

Q32 · DOK 1 · Recall

What is a goblet cell, and where is it found?

Q33 · DOK 1 · Recall

Describe the anatomy of a motile cilium from outside in, including the axoneme, dynein arms, radial spokes, and nexin links. What does each part contribute?

Q34 · DOK 1 · Recall

What does the 9+2 arrangement refer to in a cilium, and how does it differ from a 9+0 arrangement?

Q35 · DOK 1 · Recall

What is the basal body, where does it sit, what is it structurally similar to, and what does it do for the cilium?

Q36 • DOK 2 • Compare

Compare cilia and flagella by length, number per cell, motion pattern, and where each appears in the human body.

Q37 • DOK 2 • Compare

Compare motile cilia and primary (non-motile) cilia by axoneme structure, function, and one location of each.

Q38 • DOK 2 • Apply

What are stereocilia, where are they found, and how do they differ structurally from true cilia?

Q39 • DOK 3 • Reason

Primary ciliary dyskinesia (Kartagener's syndrome) is caused by defects in the dynein arms of cilia. Using what you know about ciliary structure and where cilia and flagella live in the body, predict three clinical features of someone with this condition.

Glands • 5 questions

Q40 • DOK 1 • Recall

What is a gland, and what cells form one?

Q41 • DOK 1 • Recall

Exocrine and endocrine glands differ in one defining structural feature. What is it, and how does that feature determine where each delivers its product?

Q42 • DOK 1 • Recall

Give one example of a unicellular exocrine gland and two examples of multicellular exocrine glands.

Q43 • DOK 2 • Apply

Compare merocrine, apocrine, and holocrine glands by how they release their product.

Q44 • DOK 1 • Recall

Compare serous and mucous secretions by composition and a typical site.

Connective tissue: general plan • 5 questions

Q45 • DOK 1 • Recall

List the major functions of connective tissue.

Q46 • DOK 1 • Recall

Every connective tissue is built from three ingredients. Name them, and name the term for the two non-cell ingredients together.

Q47 • DOK 2 • Apply

How does the physical state of the ground substance differ across connective tissues, and what is one example of each?

Q48 • DOK 2 • Compare

What do the suffixes -blast and -cyte mean in connective tissue cell names? Give two examples.

Q49 • DOK 1 • Recall

Name the resident cell types you would expect to find in connective tissue proper and what each one does.

Connective tissue: fibers and ground substance • 4 questions

Q50 · DOK 2 · Compare

Compare the three connective tissue fiber types by physical property and role.

Q51 · DOK 2 · Apply

Collagen is the strongest fiber. Where in the body would you predict the highest density of collagen, and why?

Q52 · DOK 1 · Recall

Name three places in the body where elastic fibers are especially abundant, and why each location needs them.

Q53 · DOK 1 · Recall

What is ground substance, what is it made of in general terms, and what does it do?

Connective tissue proper: loose and dense · 8 questions

Q54 · DOK 2 · Compare

Compare loose and dense connective tissue by how much fiber they contain and what each is built to do.

Q55 · DOK 1 · Recall

Where is areolar connective tissue found, and what does it do?

Q56 · DOK 1 · Recall

List the major functions of adipose tissue, and name two locations where it is concentrated.

Q57 · DOK 1 · Recall

Compare white and brown adipose tissue by appearance and basic role.

Q58 · DOK 1 · Recall

What is reticular connective tissue, and where is it found?

Q59 · DOK 2 · Apply

Describe dense regular connective tissue and predict its body locations from its structure.

Q60 · DOK 2 · Apply

Describe dense irregular connective tissue and predict its body locations.

Q61 · DOK 1 · Recall

Where is dense elastic connective tissue found, and why?

Cartilage and bone · 8 questions**Q62 · DOK 1 · Recall**

List the general features of cartilage as a tissue.

Q63 · DOK 2 · Compare

Compare the three cartilage types by distinguishing property and typical location.

Q64 • DOK 1 • Recall

Compare chondroblasts and chondrocytes, and explain where each is found.

Q65 • DOK 1 • Recall

What is the perichondrium, what is it made of, and what does it do?

Q66 • DOK 1 • Recall

List the general features of bone as a tissue.

Q67 • DOK 1 • Recall

Name the three main bone cell types and what each does.

Q68 • DOK 1 • Recall

Compare compact and spongy bone by structure and role.

Cartilage heals slowly; bone heals well. Explain the structural reason for this difference.

Blood is classified as a connective tissue. What feature of blood justifies that classification?

List the major functions of blood.

What is lymph, where does it come from, and where is it heading?

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Q73 · DOK 1 · Recall

What is the basic function of muscle tissue?

Q74 · DOK 2 · Compare

Compare the three muscle tissue types by striations, control, location, and cell description.

Q75 · DOK 1 · Recall

What does striated mean in muscle tissue, and which of the three muscle types are striated?

Q76 · DOK 1 · Recall

What are intercalated discs, where are they found, and what do they do?

Q77 · DOK 1 · Recall

Which muscle types are under voluntary control and which are involuntary?

Q78 · DOK 1 · Recall

Name four locations where smooth muscle is found, and one job it does at each.

Nervous tissue • 7 questions

Q79 • DOK 1 • Recall

What is the basic function of nervous tissue?

Q80 • DOK 1 • Recall

Name the two cell types of nervous tissue and state what each does.

Q81 • DOK 1 • Recall

Name the three main parts of a neuron and the role of each.

Q82 • DOK 1 • Recall

Name the glial cells of the central nervous system and what each does, in basic terms.

Q83 • DOK 1 • Recall

Name the glial cells of the peripheral nervous system and what each does.

Q84 · DOK 2 · Apply

What is myelin, what is it made from, and what does it do for the neuron it wraps?

Q85 · DOK 2 · Compare

What is the difference between a nerve and a tract?

Body membranes · 5 questions**Q86 · DOK 1 · Recall**

What is a body membrane in the anatomical sense, and what two tissue types build most of them?

Q87 · DOK 2 · Compare

Name the four body membrane types and describe where each is found.

Q88 · DOK 2 · Compare

Compare the parietal and visceral layers of a serous membrane, and explain what fills the space between them.

Q89 · DOK 2 · Apply

How is the synovial membrane different from the other three body membranes?

Q90 · DOK 1 · Recall

What are the typical functions of a mucous membrane, and what kind of epithelium often lines it?

Tissue repair · 4 questions**Q91 · DOK 1 · Recall**

Name the three general events that follow tissue injury in order.

Q92 · DOK 2 · Compare

Compare regeneration and fibrosis as outcomes of tissue repair.

Which tissues regenerate well and which scar instead?

A heart attack damages a region of cardiac muscle. Predict the structural outcome and explain its functional consequence.

A tendon must resist strong pulling forces along the line from muscle to bone. Predict which connective tissue subtype you would expect there, and justify your prediction structurally.

The urinary bladder must stretch as it fills and recoil as it empties, all without leaking urine. Predict the epithelium that lines it and the feature that suits it to that job.

Q97 · DOK 3 · Reason

Predict the epithelium that lines the alveoli of the lungs, and explain why no other type would fit.

Q98 · DOK 3 · Reason

A deep burn destroys all layers of the epidermis and exposes the underlying dermis. Using what you know about epithelium and connective tissue, explain why the wound looks raw and weeps fluid, and why re-covering it is medically urgent.

Q99 · DOK 3 · Reason

In some chronic inflammatory conditions of the airway, goblet cell numbers increase dramatically. Predict the clinical consequence using what you know about goblet cells.

Q100 · DOK 3 · Reason

An athlete tears the meniscus of the knee, a fibrocartilage structure. Predict how this injury will heal and what the long-term consequences are likely to be.

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